

Preclass Learning: Advanced Data Wrangling P1 (ICA 6)

1. Tidy Data and Reshaping

- **Tidy Data Rules:** (1) Each variable in a column. (2) Each observation in a row. (3) Each value in a cell.
- **Pivoting Longer (Gathering):** The process of turning column *headers* into data values in a single column. This converts "wide" data into the "long," tidy format required for analysis.
 - **Goal:** Use a "names_to" column for the original variable names and a "values_to" column for the associated numbers.
- **Pivoting Wider (Spreading):** The opposite process, turning the unique values of a single column into new column headers.

2. Advanced Data Cleaning

- **String Manipulation:** Working with text data (e.g., cleaning user input, extracting codes from messages).
 - **Functions:** `str_extract()`, `str_replace()`, `str_split()`.
- **Regular Expressions (Regex):** A powerful tool for sophisticated pattern matching in strings.
 - Uses special characters (`.`, `*`, `+`, `|`, `^`, `$`) to define complex search patterns.
- **Type Coercion and Parsing:** Converting data between types (e.g., a character string representing a date into a formal date object, or text numbers into numeric values).

3. Data Integrity

Ensuring data consistency and quality *before* starting any analysis. This includes standardizing capitalization, removing whitespace, and validating unique identifiers.